

Real-Time Topology & Monitoring

- Live topology mapping with bandwidth stats per link
- Dynamic alerts for link status, rogue devices, load hotspots
- Health dashboards (One-Screen) for instant network state visibility

Policy-Based Segmentation & Security

- VLAN-based network isolation
- Guest portals, client isolation, and rogue AP detection
- Switch ACLs and firewall rule management for inter-VLAN and perimeter security
- WPA3 Enterprise with RADIUS/802.1X for corporate WiFi authentication

Features for Enterprise Operations

Advanced Enterprise WiFi Features

- Band steering, fast roaming (802.11k/v/r), airtime fairness
- Mesh backhaul support for challenging deployments
- Per-SSID bandwidth controls, scheduling, QoS per SSID/VLAN
- Captive portals with voucher integration, advertising, and terms acceptance

Unified Wired and Wireless Management

- Integrated control of switch port profiles, VLAN tagging, LAG (Link Aggregation), and QoS queues
- Per-port bandwidth control and storm control
- Seamless wired/wireless segmentation for IP surveillance, VoIP, and business apps

Comprehensive Gateway Services

- Stateful firewall with NAT, DoS protection
- Dual WAN with load balancing and failover
- Site-to-Site and Client-to-Site VPN (OpenVPN/IPSec/PPTP/L2TP)
- Policy routing for traffic segmentation and inter-site access control

Omada SDN Mobile App

- Full management, monitoring, alerts, and provisioning via mobile
- Excellent for MSPs and network admins on the move
- Omada Software Controller
- Omada Software Controller is totally free, as well as all upgrades. The controller can be hosted on any computers with Windows or Linux systems on your network.

Omada Hardware Controller

Omada Hardware Controller is the management device which is pre-installed with Omada Software Controller. You just need to pay for the device, then the built-in Omada Controller software is free to use, no license fee or extra cost required. About the size of a mobile phone, the device is easy to deploy and install on your network.

Omada Cloud-Based Controller

Omada Cloud controller is deployed on the Omada Cloud server, providing paid license service with tiered pricing. With paid licenses bound to the devices on the controller, you can configure and manage the devices via Omada Cloud Service. And you need not purchase an additional hardware device or install the software on the host.

Comprehensive Omada SDN Architecture + ZTP

- Omada SDN network data/control planes
- Controller discovery mechanisms and communications
- ZTP workflows for multi-site deployments
- Omada SDN API and remote management possibilities
- Detailed security handling
- Layer 2/3 integration practices
- Advanced WiFi features management
- Omada topology visualization mechanics
- Failover, redundancy, and scalability considerations

Zero-Touch Provisioning (ZTP) for Remote and Multi-Site

ZTP is especially valuable for multi-site networks where devices ship directly to a remote branch and auto-provision once plugged in.

Steps

1. Pre-configure site profiles on Omada Controller:
 - SSID/VLAN schemes
 - AP switch port profiles
 - Gateway routing and VPN settings
2. Ship devices to site
 - Ensure network has DHCP with Option 138, or ODU installed, or cloud-bound devices
3. Device boots and discovers controller
4. Auto-adopt via controller whitelist or administrator approval
5. Controller applies configuration profile to the device
6. Device is fully operational without on-site configuration

Omada SDN integrates with larger networks via:

- 802.1Q VLAN tagging
- Inter-VLAN routing on Omada Gateways or upstream core routers
- Port trunking and LAG (Link Aggregation Groups) on switches
- VRRP not currently natively supported for gateway redundancy (yet), but load balancing/multi-WAN is.

Advanced Wi-Fi Features Management

Omada SDN handles advanced wireless operations:

- Band Steering: Push clients from 2.4GHz to 5GHz
- Load Balancing: Client cap per AP or band
- Fast Roaming (802.11k/v/r): Seamless client handoff